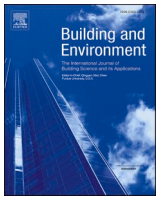
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Outdoor thermal comfort of urban plaza space under thermoacoustic interactions –Taking Datang Everbright City as an example

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ABSTRACT

The comfort of urban square environments in terms of thermal and acoustic conditions directly influences the quality of people's travel experiences and their overall well-being. This study, represented by Xi'an Datang Everbright City, investigates the impact of thermal-acoustic environments on human comfort in cold regions of China. Four typical spaces within the city square were selected for meteorological measurements and subjective questionnaire surveys to gauge the thermal perception of active crowds. Using the Universal Thermal Climate Index (UTCI) as a thermal benchmark, the study compares the interaction between thermal conditions and acoustic environments. Results indicate that the neutral UTCI in Xi'an urban squares is 14.9 °C, with a neutral UTCI range of 10.8~19.0 °C and a preferred temperature of 17.1 °C. Positive types of sounds moderately reduce thermal sensation. Moderate environmental sounds alleviate thermal discomfort to some extent among respondents. In environments such as birdsong and flowing water, people prefer higher temperatures, thereby increasing thermal tolerance. Conversely, in noise from traffic or construction, individuals prefer lower temperatures, reducing their thermal tolerance. In environments with broadcasts or conversations, respondents exhibit slightly higher thermal tolerance compared to the average neutral temperature. Finally, based on meteorological characteristics at different survey points and respondent feedback, optimization strategies for design are proposed.

1. Introduction

With the rapid development of cities, China's urbanization process has accelerated significantly, leading to substantial changes in urban meteorological conditions [1]. According to data from the National Bureau of Statistics in 2022, China's urbanization rate has increased dramatically from 17.9 % in 1978 to 64.72 % in 2021 [2], and the urban resident population has grown from 170 million to 910 million Xi'an, for instance, has an urban population of 10.34 million and an urbanization rate of 79.59 % [3], making it a representative example of a highly urbanized city. While this rapid urbanization reflects the progress of China's modern economic development, it also imposes considerable pressure on the urban ecological environment.

The urban heat island effect is one of the environmental issues resulting from rapid urbanization. Modern cities have seen an increasing number of buildings and structures that absorb significant amounts of solar radiation during the day and release this heat at night, causing a noticeable increase in temperatures in urban centers. Additionally, the urban heat island effect exacerbates temperature differences within the city. This temperature variation has widespread impacts on both residents and the environment [4]. Excessive temperatures not only cause health issues for residents, such as heatstroke and respiratory difficulties, but also lead to increased energy consumption, particularly due to the higher use of air conditioning.

Therefore, mitigating the urban heat island effect and enhancing environmental comfort in cities have become significant topics of research and discussion among scholars in the field of the built environment. For instance, Negar et al. [5] used measurements and questionnaires to explore outdoor thermal comfort dimensions on acoustic comfort, emphasizing plant communities in urban parks, to derive metrics of thermal comfort in relation to landscape quality. Li et al. [6]

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Nomenclature

LS	Leisure Space
OS	Open Square
PS	Pedestrian Street
SS	Space under Shadow
OTC	Outdoor thermal comfort
T_a	Air temperature
RH	Relative humidity
V_a	Wind speed
T_g	Globe temperature
T_{mrt}	Mean radiant temperature
G	Global radiation
TSV	Thermal sensation vote
MTSV	Mean thermal sensation vote
TCV	Thermal comfort vote
MTCV	Mean thermal comfort vote
UTCI	Universal thermal climate index
NUTCI	Neutral universal thermal climate index
NUTCIR	Neutral universal thermal climate index range
PPD	Predicted percentage of dissatisfied
ASV	Acoustic sensation vote
MASV	Mean acoustic sensation vote
ACV	Acoustic comfort vote
MACV	Mean acoustic comfort vote
SPL	Sound pressure level

proposed 13 practical greenway design strategies to enhance thermal comfort during greenway walking by taking into account the physical environment, walking condition and psychological factors through meteorological measurements and questionnaire surveys. Xue et al. [7] found that bodies of water can significantly reduce the temperature of surrounding open spaces, thereby alleviating the heat island effect.

Urban squares, as typical public spaces within cities, provide areas for residents and tourists to rest and interact. Therefore, it is essential to mitigate the heat island effect in these squares and enhance their thermal comfort. However, in real environments, the factors affecting human comfort are complex, and studies focusing solely on the thermal environment cannot comprehensively reflect the level of comfort. Additionally, with advancements in technology and the continuous innovation of research methods, studies considering multiple factors under composite conditions have gradually emerged in recent years. Gao et al. [8] analyzed the weight factors of various influencing factors in the outdoor environment of urban residential areas and established an evaluation method for thermal and acoustic environments. Ali et al. [9] investigated gender differences in acoustic-thermal comfort and cognitive performance in several simulated work environments with load levels, and showed that gender differences in acoustic-thermal comfort and performance were nearly significant, emphasizing the importance of considering gender differences when designing indoor work environments for optimal comfort and performance. Du et al. [10] studied Multiple effects of visual-acoustic-thermal perception on the overall comfort of older adults in residential outdoor environments, resulting in a regression equation that predicts the overall comfort of older adults. Ying et al. [11] used a university campus as a research site to explore the changing characteristics of human comfort in thermo-acoustic environments, and concluded that both acoustic and thermal environments are important factors affecting subjective human comfort, but thermal environments are more important than acoustic environments. Yin et al. [12] explored the impact of background music on human comfort and work efficiency, concluding that changes in environmental parameters can affect psychological emotions and physiological parameters, altering levels of human comfort. Xiao et al. [13] investigated the impact of environmental quality on thermal perception under different visual and auditory conditions, demonstrating a negative correlation between overall comfort and environmental parameters, provides research support for urban space planning and design, and facilitates the creation of better outdoor environments. Wen et al. [14] simulated different levels of thermal and acoustic environments and office activity levels through climate chamber experiments, The conclusion that thermal environments have little effect on comfort at rest, but a significant impact during reading, guides the thermal and acoustic design of office building spaces. Yang et al. [15] studied the interaction of thermal and acoustic environments on indoor thermal comfort at Terminal 3 of Xi'an Xianyang Airport and proposed optimization strategies. Zhen et al. [16] researched campus spaces, demonstrating that in most locations, thermal evaluation is primarily influenced by PET, while acoustic evaluation is mainly affected by LAeq. The interaction of thermal and acoustic environments affects people's overall comfort, however, focusing on thermal environments alone does not provide a comprehensive picture of human comfort levels. Multifactorial comfort studies under composite conditions have gradually emerged in recent years. Zhang et al. [17] explored the effects of four dimensions of soundscape on acoustic comfort in open public spaces, and the results showed that the dominance of the sound source had a significant effect on relaxation, communication, spatial sense and dynamics.

Most current research focuses on the interaction of various comfort factors in laboratory, indoor, or semi-open spaces, The results show that under thermoacoustic interaction conditions, factors such as sound can significantly affect human sensitivity to the thermal environment; however, existing indoor thermal comfort studies tend to measure under stable thermoacoustic conditions, where the thermoacoustic parameters are relatively stable with respect to the site environment, and it is not possible to accurately integrate a wide range of parameters. Meanwhile, outdoor thermoacoustic interaction study sites have complex thermoacoustic conditions, and existing studies are less likely to quantify the type of acoustic environment according to the subjective perception of respondents. Existing studies have shown that improving the layout and facilities of the plaza space can increase people's thermal-acoustic comfort to a certain extent [18]. This study investigated the interaction between human acoustic perception and thermal perception in scenic plaza spaces through meteorological parameter measurements and questionnaire distribution, combined with statistical analysis of subjective polling, and used this as a basis to provide design optimization strategies for outdoor environments, providing new ideas and directions for future urban plaza design and management.

2. Methods

2.1. Study area

Xi'an is located in northwestern China (34°16′N 108°54′E / 34.267°N 108.9°E). According to the Köppen climate classification, Xi'an lies at the junction of a cold semi-arid climate and a humid subtropical climate [19], characterized by a temperate continental monsoon climate with distinct four seasons. Summers are hot and humid, while winters are cold and dry. Based on annual temperature data from 2010 to 2021, the highest monthly average temperature (Ta) occurs in July (28.08 °C), with a maximum Ta of 39.33 °C. The lowest monthly average Ta is in January (0.62 °C), with a minimum Ta of −8.01 °C. The annual average relative humidity ranges from 51.28 % to 73.68 %. According to China's current climate zoning regulations [20], Xi'an is classified as a cold region with long, cold, and dry winters, presenting significant opportunities for outdoor environment enhancement. To investigate the impact of thermal and acoustic environments on human comfort in

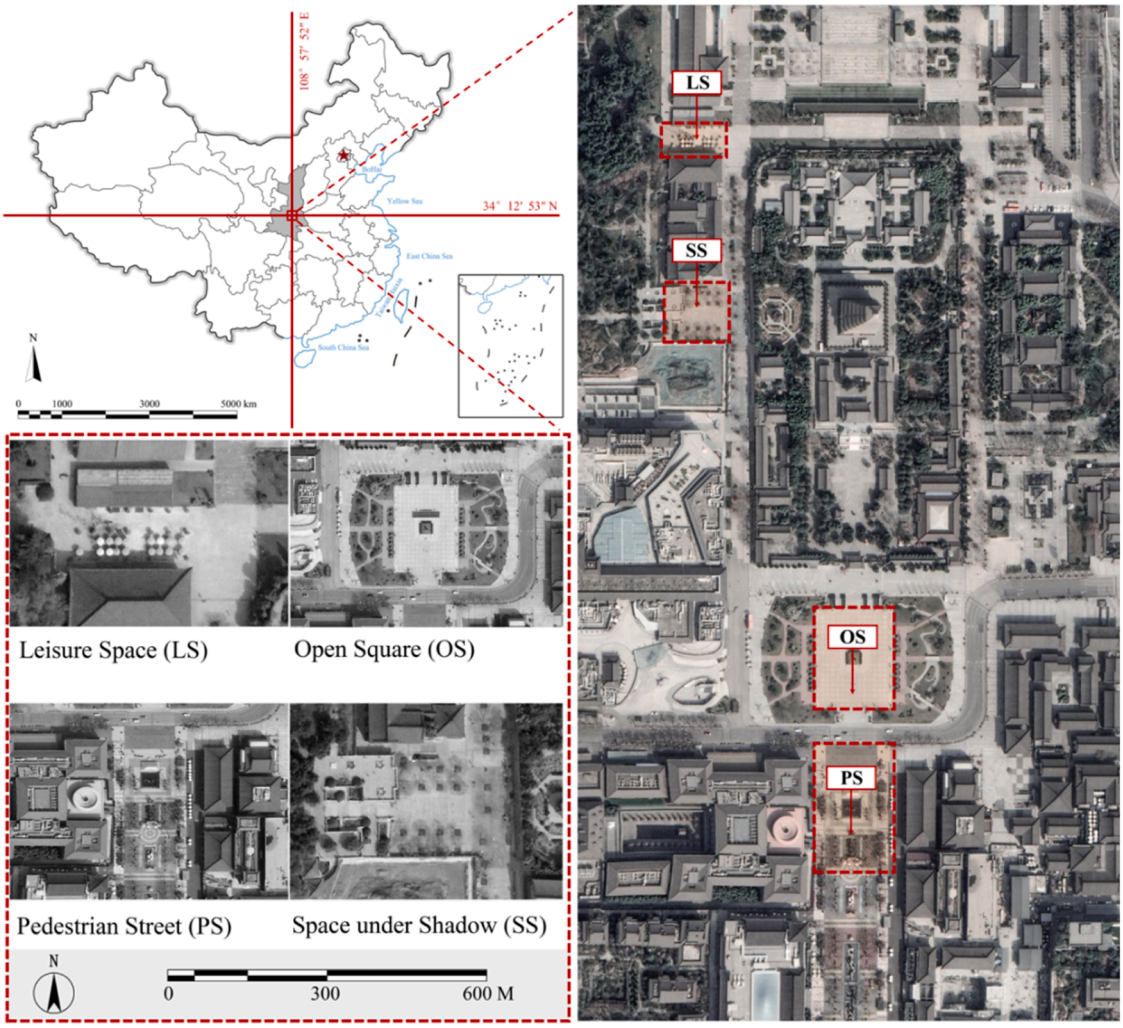


Fig. 1. Site locations.

square spaces, this study selects the front square of the bustling tourist area, the "Datang Everbright City" (34°13′5″N, 108°57′51″E), as the research location (Fig. 1).

Datang Everbright City, chosen as the study area for this research, features a relatively flat and open terrain adorned with architectural complexes reminiscent of China's Tang Dynasty. It includes water features and landscaped gardens. The main street runs north to south for 2100 m and east to west for 500 m, covering a total construction area of 650,000 m^2s. The front square is rectangular with a high enclosure degree and lacks tall structures within the site. Based on field surveys, this study selects four representative open space nodes: LS measurement point is a shaded resting area covered by tree canopies. It is enclosed by brick and stone walls ranging from 2.4 to 4 m in height on the north and south sides. There are tall deciduous trees on the southern side, and the predominant sound is pedestrian conversation. OS measurement point is an open square space located along the central axis. It is predominantly paved with marble, surrounded by greenery on all sides. PS measurement point is a street-side pedestrian walkway adorned with tree shade and water features. The street is lined with trees approximately 10 m tall on both sides. To the east of the measurement point, there is a landscaped pool with a depth of approximately 0.5 m. The overall sound environment is complex. SS measurement point is a shaded resting area with ample outdoor sunshades. To the south, there is a five-story shopping mall, and the greenery mainly consists of small shrubs.

2.2. Experimental design

The comprehensive evaluation of human thermal comfort in varying environmental conditions is influenced by multiple meteorological parameters [21]. In this experiment conducted outdoors, all meteorological parameters relevant to human thermal sensation were measured. The research period for this study lasted from September 2022 to September 2023, a one-year period covering the transition season, winter and summer. In Xi'an, temperatures are significantly lower in the early morning and evening compared to daytime, resulting in large diurnal temperature variations. Nearly all residents of Xi'an for over a year mitigate the impact of these temperature differences on comfort by adjusting clothing and outdoor activities. Therefore, the research was conducted between 9:00 and 17:00 to avoid these times.

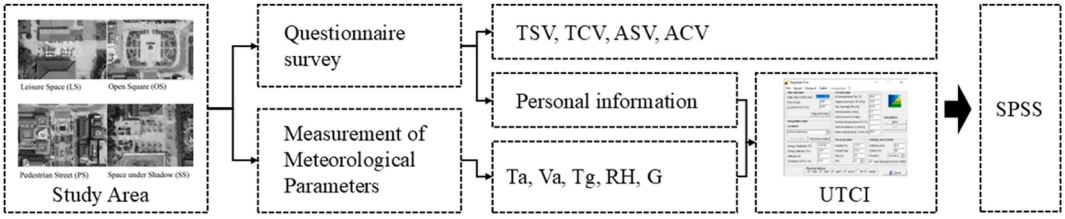
Furthermore, rainy or snowy weather not only affects the heat exchange of ground surfaces in open squares but also influences respondents' subjective choices in voting. Consequently, this study selected days with suitable weather conditions such as clear skies or light cloud cover, conducive to outdoor activities for the survey.

2.2.1. Measurement of meteorological parameters

In each of the four designated measurement zones, instruments were installed to monitor meteorological data (Fig. 2(a)). The monitored meteorological parameters include air temperature (Ta), relative humidity (RH), wind speed (Va), solar radiation (G), globe temperature (Tg), and sound pressure level (dB) (Table 1). Globe temperature (Tg)



(a)



(b)

Fig. 2. (a) Installation of measuring instruments (b)Research methodology.

Table 1
Instruments used for acquiring meteorological variables and sound pressure level recording.

Instrument	Parameter	Range	Accuracy
JT2020	Air temperature	0–50 °C	±0.6 °C
	Relative humidity	0–100 %	±3 %
	Globe temperature	0–80 °C	±1 °C
TES1333	Global radiation	2000 W/m^2	0.1 W/m^2
JT2020	Wind speed	0.3–45m/s	0.001 m/s
TES-1350A	Sound pressure level	A LO(Low) 35–100 dB	0.1 dB

represents the comprehensive environmental sensation temperature experienced by humans due to radiant heat and convective effects, incorporating factors such as ambient air temperature and radiation. It reflects the thermal conditions felt by the human body from radiant heat and strongly influences mean radiant temperature (T_{mrt}), which is calculated according to Equation (1.1) as per ISO 7726 standard [22]:

$$T_{mrt} = \left[(T_g + 273)^4 + \frac{1.10 \times 10^8 V_a^{0.6}}{\varepsilon D^{0.4}} (T_g - T_a) \right]^{\frac{1}{4}} - 273 \quad (1.1)$$

Here, D represents the diameter of the black globe (in this study, $D = 0.05$ m), and ε denotes the emissivity (for the black globe, ε=0.95).

Before the experiment, the instruments were calibrated and fixed on a stand 1.5 m above the ground and measurements of meteorological parameters will be taken every minute. In measuring the acoustic environment, given the instability of ambient sound pressure levels under natural conditions, we recorded the average sound pressure level of the environment during the respondents' questionnaire sessions to minimize the error caused by instantaneous noise. Additionally, the surveyors noted the type of sound present at the time: traffic, construction, birdsong, music, broadcasting, or conversation.

2.2.2. *Questionnaire survey*

In this experiment, a questionnaire survey method was used to collect data from respondents. Randomly selected respondents at the measurement points were given questionnaires. While the respondents filled out the questionnaires, the surveyors simultaneously measured meteorological data to ensure the precision and timeliness of the data. The survey was conducted on days suitable for outdoor activities, such as sunny or cloudy days, avoiding uncomfortable weather conditions like rain, snow, and sandstorms, which could lead to abnormal meteorological parameters.

The questionnaire consisted of two main parts (Specific questionnaire information can be found in the appendix). The first part collected basic information about the respondents, including age, gender, length of residence in Xi'an, and physical activity status in the past 20 min. Due to the differences in the perception of thermal sound, and the degree of thermal sound sensitivity of respondents in different age groups, in the questionnaire survey, in order to avoid such errors, we divided the respondents according to the age interval they were in, and limited the number of respondents in each interval to an appropriate level. The metabolic activities were referenced according to the standard metabolic rates outlined. The second part investigated respondents' perceptions of the thermal and acoustic environment in public open spaces, including Thermal Sensation Vote (TSV), Thermal Comfort Vote (TCV), thermal preference, thermal acceptability, Acoustic Sensation Vote (ASV), Acoustic Comfort Vote (ACV), acoustic preference, acoustic acceptability, and overall comfort. At the same time, the main types of environmental sounds perceived by the respondents during the questionnaire filling process (insects, bird calls, flowing, traffic, machines, hawking, radio, music and crowds) were recorded and categorized after the data collection was completed.

To quantify different levels of thermal and acoustic perception and comfort, this study used the Likert scale to reflect respondents' perceptions of the current thermal and acoustic environment. According to the ASHRAE definition, a 7-point Likert scale was used in the questionnaire for thermal sensation (−3, cold; −2, cool; −1, slightly cool; 0,

Table 2
Personal information of respondents in surveys.

Location		Total	LS	OS	PS	SS
Research period	Transition season	299	16	92	70	121
	Winter	281	22	116	80	63
	Summer	424	194	71	56	103
Biological sex	Male	571	124	182	101	164
	Female	433	108	97	105	123
Age	≤18 years	119	50	30	20	19
	19–25 years	430	79	120	94	137
	26–40 years	345	67	104	69	105
	41–55 years	84	25	23	15	21
	>55 years	26	11	2	8	5
Time living in Xi'an	0–6 Months	598	153	184	113	148
	6 Months - 1 Year	37	8	9	11	9
	More than 1 year	369	71	86	82	130

Table 3
Meteorological parameters and heat indices calculated for each study site. SD: standard deviation.

Space		LS	OS	PS	SS
Air temperature (°C)	Max	40.30	40.7	40.2	40.1
	Min	12.50	8.0	10.1	11.3
	Mean ± SD	32.6 ± 7.4	23.31± 9.21	22.02 ± 9.33	25.82± 8.45
Relative humidity (%)	Max	58.8	52.3	58.9	53.3
	Min	14.00	10.3	16.3	10.2
	Mean ± SD	33.6 ± 9.2	26.20 ± 10.98	37.07 ± 8.97	31.06 ± 10.75
Wind speed (m/s)	Max	2.15	2.17	2.15	1.55
	Min	0.4	0.4	0.4	0.2
	Mean ± SD	0.56 ± 0.40	0.63 ± 0.35	0.54 ± 0.30	0.53 ±0.22
Globe temperature (°C)	Max	41.8	45.7	41.2	41.2
	Min	15.9	8.5	11.4	11.2
	Mean ± SD	33.7 ± 7.3	25.89 ± 9.97	23.2 ± 9.37	27.08 ± 8.33
Global radiation (W/m2)	Max	189.20	994.9	947.2	937.1
	Min	9.10	23.3	16.2	19.4
	Mean ± SD	70.89 ±31.05	413.16 ±273.83	161.23 ±191.49	132.25 ±128.87
Sound pressure level (dB)	Max	69.6	77.6	74.3	73.9
	Min	47.5	47.7	46.4	40.7
	Mean ± SD	58.00 ± 4.22	63.26 ± 7.02	60.02 ± 5.81	59.29 ± 6.92

neutral; 1, slightly warm; 2, warm; 3, hot), thermal comfort (–3, very discomfort; –2, discomfort; –1, slightly discomfort; 0, neutral; +1, slightly comfort; +2, comfort; +3, very comfort), acoustic sensation (–3, very loud; –2, loud; –1, slightly loud; 0, moderate; +1, slightly quiet; +2, quiet; +3, very quiet), and acoustic comfort (–3, very discomfort; –2, discomfort; –1, slightly discomfort; 0, neutral; +1, slightly comfort; +2, comfort; +3, very comfort). For evaluating preferences for meteorological factors, a 3-point scale was used (−1, decrease/weaker; 0, unchanged; 1, increase/stronger). Additionally, respondents' cooling measures adopted during the summer were surveyed to determine their thermal adaptation behaviors.

During the questionnaire completion process, the experimenters assisted the respondents in understanding the content of the survey to help them fill it out more accurately, thus closely reflecting the respondents' true feelings. A total of 1153 questionnaires were collected from four measurement points, of which 1004 were valid.

2.2.3. Thermal index

In this study, the Universal Thermal Climate Index (UTCI) was selected as the thermal index. This index was developed by the International Society of Biometeorology (ISB) based on the Fiala multi-node thermoregulation model and the adaptive clothing model [23]. The UTCI is a comprehensive thermal comfort evaluation index widely used in existing research. It takes into account human metabolic processes, heat accumulation, and heat transfer, making it the most thoroughly considered and deeply calculated thermal index. The UTCI can be used for thermal comfort model predictions across different climate zones, environmental characteristics, and populations of different races. The calculation of this index incorporates various thermal environment parameters such as temperature and humidity and is typically performed using the RayMan software. Fig 2(b) illustrates the overall process of conducting the empirical tests and questionnaires in this study.

3. Results

3.1. Descriptive analysis

3.1.1. Respondent attributes

In this study, a total of 1153 questionnaires were collected, with 1004 valid responses. Among these, 232 were from the LS site, 279 from the OS site, 206 from the PS site, and 289 from the SS site (Table 2). The survey period included dates with significant temperature variations, covering winter, summer, and transitional seasons. Table 2 presents the personal information of the respondents surveyed during the study. Among the respondents, 43.1 % were female and 56.9 % were male, indicating a higher proportion of males. The demographic at the Datang Everbright City showed a tendency towards youthfulness, with the majority of respondents aged 19–25 years, accounting for 42.8 % of the total survey population; followed by those aged 26–40 years, comprising 34.4 % of the surveyed group. A significant portion of respondents had lived in Xi'an for less than 6 months (59.6 %), suggesting that many respondents were recent tourists to Xi'an, likely drawn by the attractions of Datang Everbright City.

3.1.2. Meteorological parameters

Through descriptive analysis using SPSS software, the maximum, minimum, mean, and standard deviation of meteorological parameters at four measurement points were computed (Table 3). It was observed that the OS site had the highest solar radiation (994.9 W/m^2) and black globe temperature (45.7 °C), likely due to the absence of surrounding buildings or trees and the reflective surface of marble flooring absorbing significant heat. Consequently, this site also recorded the highest average wind speed (0.63 m/s) and lowest air humidity (26.2 %). In contrast, the LS site exhibited the lowest average solar radiation (70.89 W/m^2), while the PS and SS sites showed similar solar radiation levels (161.23 W/m^2 and 132.25 W/m^2, respectively). Observations indicated that the LS site was predominantly shaded by buildings, whereas the SS and PS sites were semi-open spaces enclosed on their eastern and western sides. Additionally, the PS site had the highest air humidity (58.9 %) likely due to a nearby water feature causing increased relative humidity through evaporation. The LS site showed significantly higher average temperatures and black globe temperatures compared to the other three locations (Ta: 32.6 °C, Tg: 33.7 °C). The SS site recorded the lowest average wind speed (0.53 m/s), attributed to wind obstruction by high walls and trees on both sides. Lastly, the PS site had the lowest average temperature (22.02 °C), possibly influenced by heat absorption by the water feature to some extent.

The sound level meter data indicates that there is minimal variation in the acoustic environment between the SS, PS, and OS measurement points. Analyzing the results of the questionnaire in conjunction with the questionnaire, we found that the negative sound within the OS space and the PS space maintained a high level. Among the four points, the OS site recorded the highest noise pressure level and maximum average

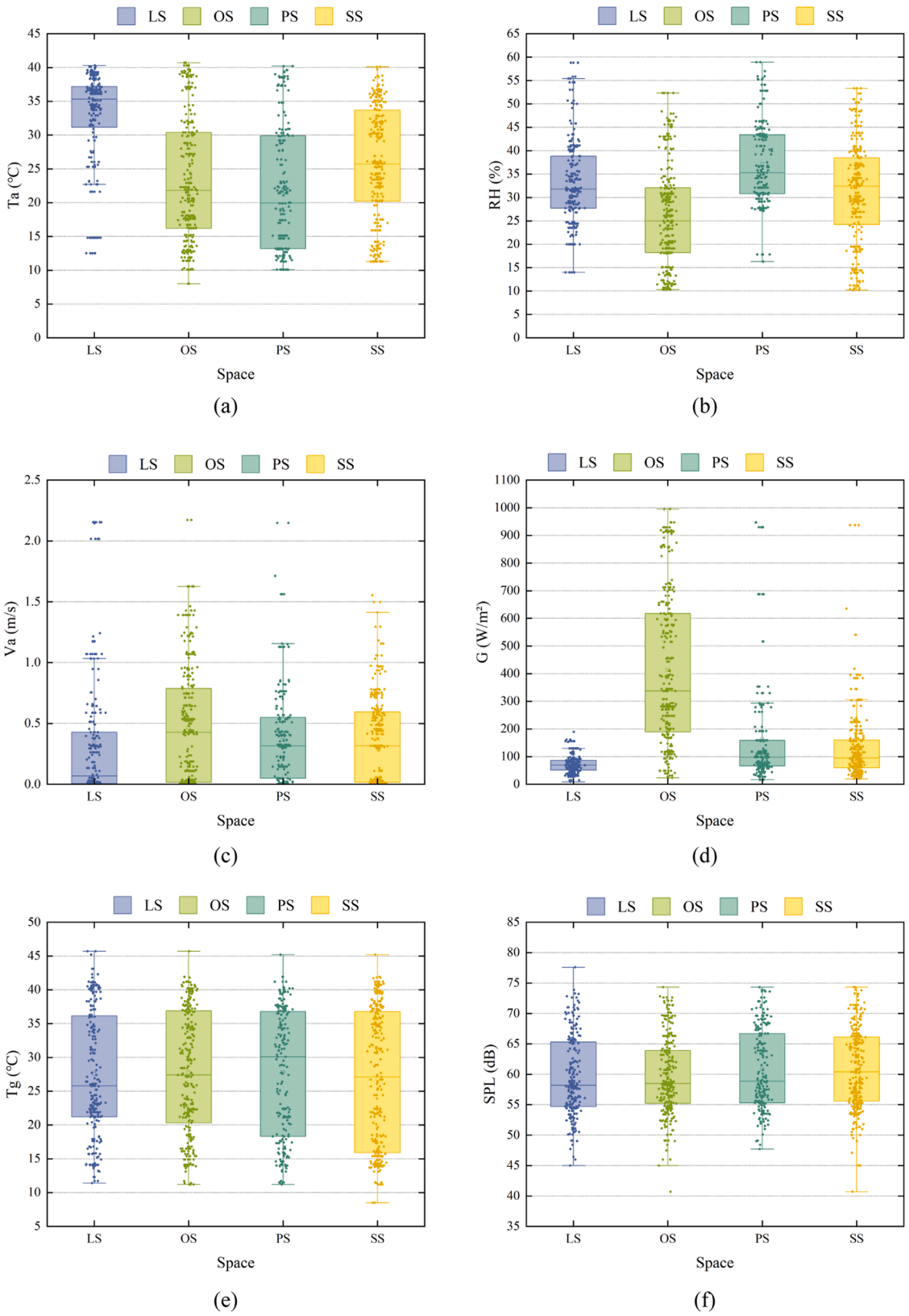


Fig. 3. Distribution of meteorological parameters: Air temperature(a), Relative humidity(b) , Wind speed(c), Global radiation(d) , Globe temperature(e), Sound pressure level(f).

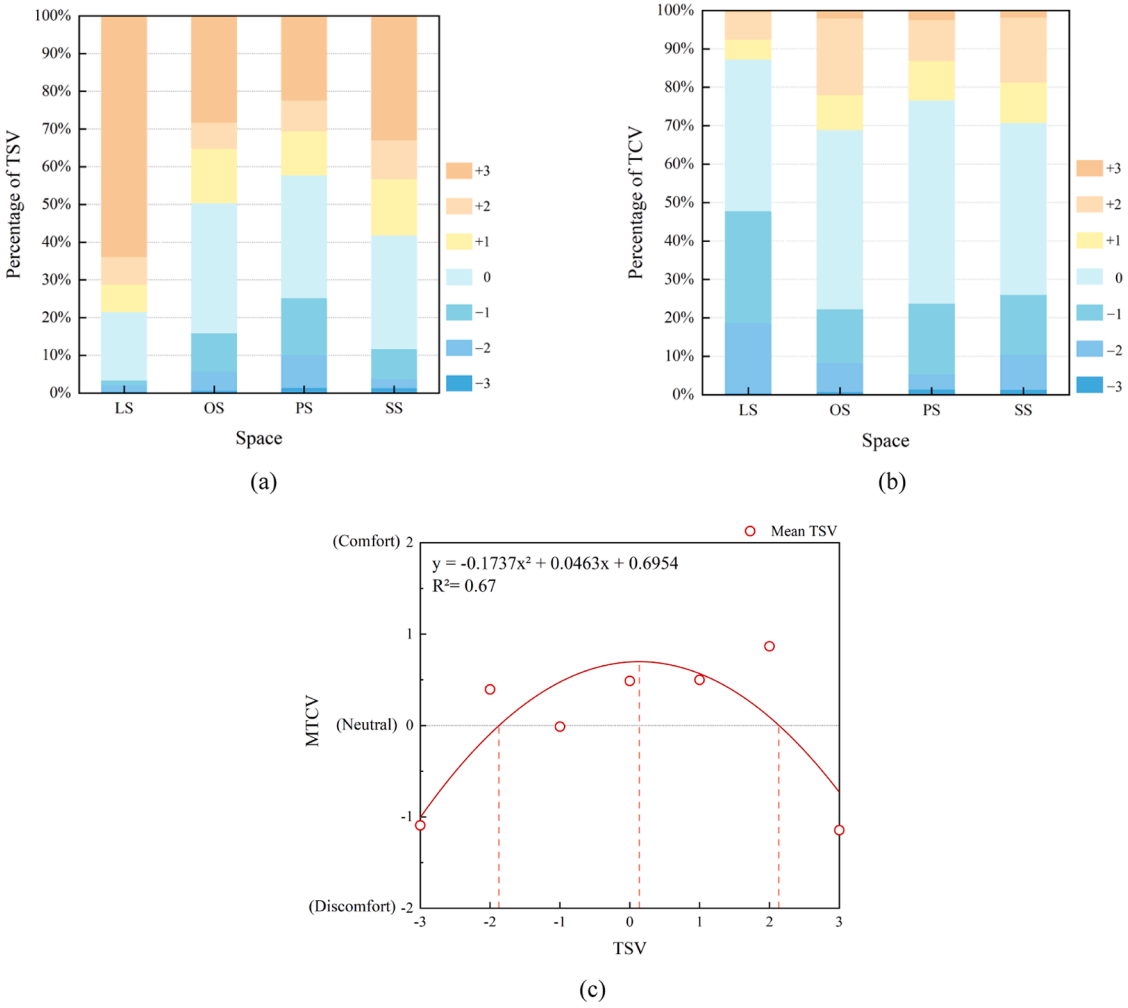


Fig 4. (a) TSV distribution, (b) TCV distribution, (c) Correlation between TSV and TCV.

noise (77.60 dB and 63.26 dB, respectively). In contrast, the LS site exhibited the lowest average noise level, measuring only 58.00 dB, indicating a relatively stable acoustic environment. Field analysis revealed that the LS site is an isolated activity area, away from pedestrian flow and road traffic, contributing to its quieter acoustic environment. In contrast, the other three locations are situated along the central axis of Datang Everbright City, frequently experiencing high pedestrian traffic, resulting in unavoidable noisy conditions.

Fig 3 depicts the specific distribution of meteorological parameters within each measurement point, reflecting their typical characteristics distinctly. It is evident that the OS site exhibits the widest range of air temperatures. The LS site, characterized by long-term shade and minimal solar radiation, maintains the most stable temperature distribution and the lowest average temperature. Regarding average humidity, the order from highest to lowest is PS (37.07 %), LS (33.60 %), SS (31.06 %), and OS (26.2 %). The SS and LS sites show similar air humidity levels. The presence of water features at the PS site significantly increases air humidity, while the OS site, with its single reflective surface, experiences markedly lower humidity levels. Solar radiation at the LS site remains stable due to its shaded location, resulting in minimal distribution of solar radiation within the area. Conversely, the OS site, being unobstructed, experiences the highest variability in solar radiation over time, with levels consistently higher than other areas. The layout differences between the PS and SS sites result in minor variations in solar radiation. There are no significant differences in average wind speed across the four measurement points during the survey period, indicating a stable wind environment in the plaza space.

3.1.3. *Thermal sensation vote and thermal comfort vote*

Fig 4(a) illustrates the distribution of Thermal Sensation Votes (TSV) during the survey period. Overall, the majority of respondents voted as feeling "very hot" (TSV=3), accounting for 41.3 %. The next most frequent sensation was "neutral" (TSV=0), while very few respondents felt "very cold" (TSV=−3), comprising only 3.7 %. At the LS site, the highest proportion of respondents selected "very hot" (TSV=+3), totaling 63.7 %, significantly more than the other three locations. This perception of warmth at the LS site is largely due to the majority of votes being collected during the summer months, as indicated by the survey distribution dates in Table 2. The distribution of Thermal Sensation Votes at the OS and SS sites was similar, while the PS site generally felt cooler overall. This is attributed to landscape features such as water pools in the area, which lower local temperatures.

Fig 4(b) depicts the distribution of Thermal Comfort Votes (TCV) across the four spaces during the survey period. It shows that the majority of respondents (70.2 %) felt comfortable with the current thermal environment, giving votes ranging from "neutral" (TCV=0) to "slightly comfortable" (TCV=+1), "comfortable" (TCV=+2), and "very comfortable" (TCV=+3). Within the LS space, the number of people who felt comfortable (TCV = 0 to +3) dropped significantly due to the large number of TSV votes for (hot, +3). Even so, a higher number of respondents felt comfortable, suggesting that people in Xi'an had a higher tolerance for hotter environments during the research period. A small percentage (5.1 %) felt "very comfortable" (TCV=+3), indicating generally favorable thermal conditions across the four measurement points. At the OS site, the highest proportion of votes (77.3 %) expressed

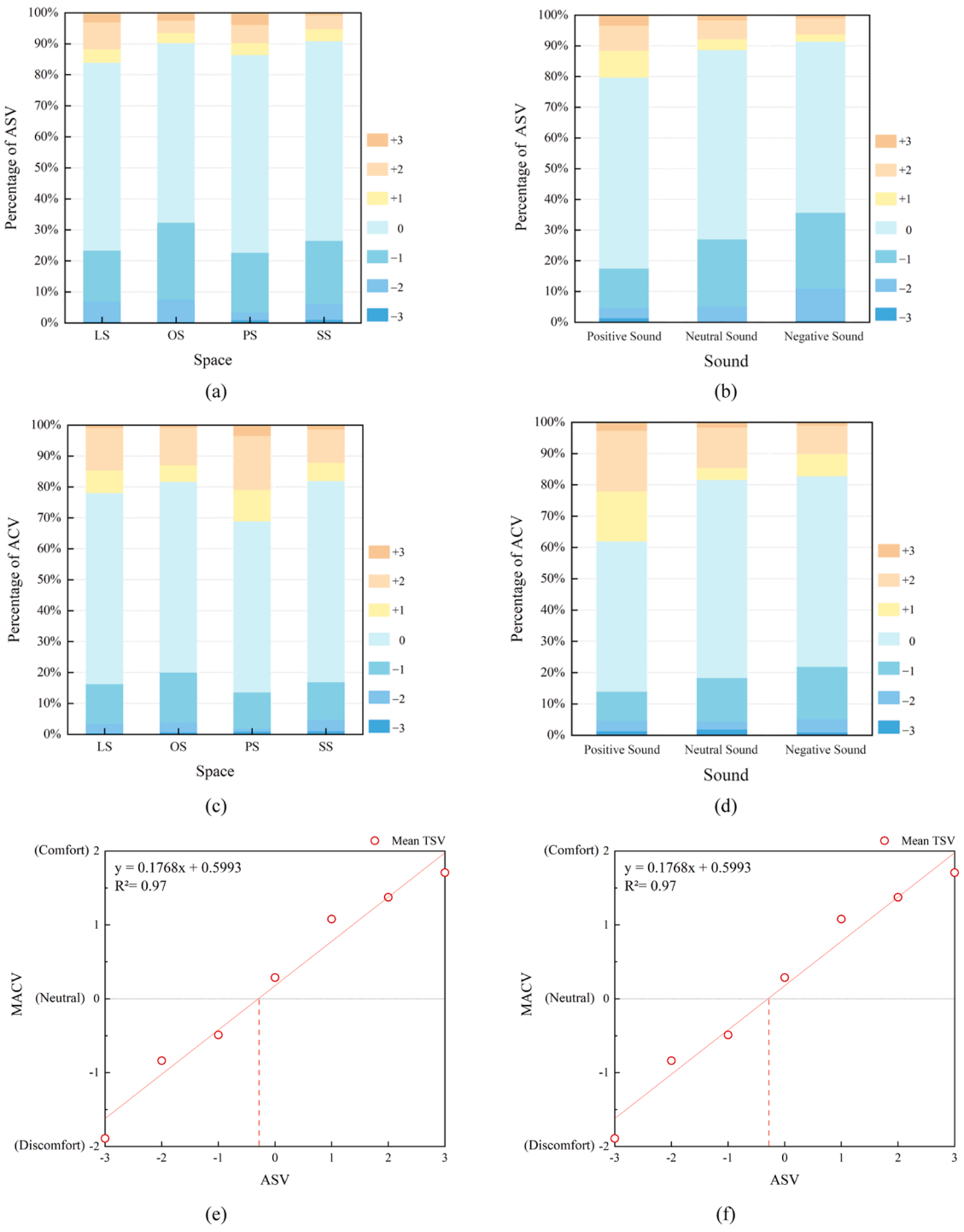


Fig. 5. (a) ASV for different spaces,(b) ASV for different sound types, (c) ACV for different spaces, (d) ACV for different sound types, (e) Correlation between ASV and ACV, (f) Correlation between ASV and ACV for different sound types.

satisfaction with the current thermal environment, consistent with the distribution of Thermal Sensation Votes shown in Fig 4(a). This indicates that the OS site had the most respondents (61.1 %) providing moderately positive evaluations (TCV=0 to +3) of the current thermal environment. In contrast, the LS site received the poorest Thermal Comfort Vote, with a significant majority (73.8 %) finding the current environment uncomfortable (TCV=−1 to −3). This proportion is much higher compared to other spaces, with only 13.7 % of respondents considering the thermal environment somewhat comfortable (TCV=0 to +3).

Each value of Thermal Sensation Vote (TSV) was averaged across all Thermal Comfort Votes (TCV), yielding the Mean Thermal Comfort Vote (MTCV). These values were plotted on the same graph and subjected to nonlinear quadratic fitting to produce Fig 4(c). It depicts the relationship between human thermal sensation and thermal comfort during the survey period. The results indicate that when TSV=0.18, people experience the highest thermal comfort (TCV=0.74), indicating a preference for moderately warm temperatures. When TSV ranges between −1.89 (cold) and 2.11 (hot), people feel comfortable with the current thermal environment (TCV > 0).

Table 4
Spearman's correlation coefficients between TSV and physical variables.

Space	Air temperature		Relative humidity		Wind speed		Global radiation	
	P	r	P	r	P	r	P	R
LS	0.000	0.674**	0.754	0.021	0.417	0.054	0.350	0.062
OS	0.000	0.590**	0.016	0.145	0.389	−0.052	0.000	0.321**
PS	0.000	0.584**	0.018	0.166	0.344	−0.067	0.017	0.168
SS	0.000	0.633**	0.240	−0.070	0.017	−0.142*	0.003	0.176*

P: **Significantly different at $P < 0.01$ (two-tailed test). *Significantly different at $P < 0.05$ (two-tailed test).

3.1.4. Acoustic sensation vote and acoustic comfort vote

In order to simplify the types of acoustic environments and investigate their impact on human comfort, this study categorized the sounds into three different types based on respondents' perceptions: positive sound sources (insect sounds, natural wind sounds, bird songs, and flowing water sounds), neutral sound sources (vendors' calls, broadcast voices, music, and crowd sounds), and negative sound sources (traffic noise and machinery sounds).

The distribution of acoustic satisfaction across measurement points is illustrated in Fig 5(a), (b). The majority of respondents (73.3 %) rated the current acoustic environment as moderate (ASV=0) or satisfactory (ASV=2.3). There were significantly more respondents perceiving the current acoustic environment as noisy (ASV<0) compared to those perceiving it as quiet (ASV>0), indicating generally lower acoustic environment quality across the four spaces. Respondents at OS and SS sites rated the acoustic environment lower than other spaces, likely due to their proximity to open squares near roads and tree-shaded relaxation areas with high pedestrian traffic, leading to poorer acoustic environments.

The distribution of acoustic comfort across measurement points is depicted in Fig 5(c), (d), indicating that the majority (83.5 %) of respondents felt comfortable (ACV≥0) with the current acoustic environment. Specifically, the highest level of acoustic comfort was reported at the PS site, with the highest proportion (31.9 %) of respondents feeling comfortable (ACV≥0) in the current environment. Respondents at OS and SS sites rated the acoustic environment less comfortable (ACV<0) (16.2 % and 14.3 %, respectively) compared to PS and LS sites (18.1 % and 19.8 %, respectively), consistent with the findings in Fig 5 (a), (b). This suggests that the noisy acoustic environments at OS and SS sites contribute to a lower overall level of acoustic comfort.

Similar to the acoustic sensation vote, the proportion of votes indicating comfort with the acoustic environment gradually decreases as the acoustic environment shifts from positive sound types to negative ones. When the acoustic environment includes sounds such as bird songs and flowing water, the proportion of respondents feeling comfortable reaches its peak at 38.6 %, significantly higher than the other two acoustic types. Only 13.9 % of respondents find the acoustic environment uncomfortable. This indicates that positive acoustic environments significantly enhance people's comfort with sound, highlighting higher tolerance for positive environmental sounds.

To explore the relationship between human acoustic perception and acoustic comfort during the survey period, we normalized all votes for Acoustic Comfort Votes (ACV) and calculated the Mean Acoustic Comfort Votes (MACV). The resulting MACV values were plotted on the same graph for linear regression analysis, yielding Fig 5(e). This figure depicts how perceived acoustic environments influence acoustic comfort. Overall, participants rated their acoustic comfort highest in quieter surroundings, with acoustic comfort decreasing as perceived noise levels increased. Specifically, when ACV = 0, ASV = −0.27, indicating moderate acoustic comfort assessment in relatively noisy environments (ASV = −0.27).

To investigate the impact of different acoustic environments on acoustic comfort, we examined variations in the relationship between acoustic perception and acoustic comfort across three types of acoustic environments. These were plotted together on a single coordinate system, each subjected to linear regression analysis to derive patterns of acoustic comfort variation (Fig 5(f)). The results show that for positive sound types, the slope of acoustic comfort change with acoustic perception is lowest, followed by neutral sound types, while negative sound types exhibit the highest slope. Moreover, under constant acoustic perception voting, respondents reported lower acoustic comfort in negative noise environments compared to neutral noise environments, with positive noise environments yielding the highest acoustic comfort votes. These findings indicate that in negative noise conditions, changes in acoustic perception significantly affect acoustic comfort; furthermore, increasing sound pressure levels further decrease acoustic comfort relative to the other two acoustic environment conditions.

3.1.5. Relationship between thermal sensation and meteorological parameters

When individuals are in outdoor spaces, meteorological parameters such as air temperature and solar radiation significantly impact their thermal perception [24]. Variations in these parameters lead to differences in subjective evaluations of the thermal environment. To explore the effects of four factors on thermal sensation in different spaces, partial correlation analysis was conducted between Thermal Sensation Vote (TSV) and four key meteorological parameters, as shown in Table 4. In analyzing the correlation between a specific meteorological parameter and TSV, the other three parameters were controlled, thereby isolating the influence of each parameter on thermal perception. The r value represents the correlation coefficient between thermal sensation and the meteorological parameter, indicating the parameter's influence on TSV, while P signifies the significance of the parameter's correlation.

The results indicate that air temperature is the most critical factor affecting thermal comfort across all spaces ($P \leq 0.01$), positively correlating with TSV, which aligns with subjective perceptions. Relative humidity and wind speed were found to be the least significant factors, showing weak correlations with thermal sensation and having no significant impact at most measurement points. In OS site, solar radiation exhibited the strongest correlation with thermal sensation ($P \leq 0.001$), followed by SS site, consistent with environmental factors. The OS site, being open squares with no surrounding obstructions, did not block solar radiation, resulting in higher G values and increased TSV. Conversely, in LS site, located in shaded areas and not exposed to direct sunlight during the survey period, the correlation was weakest ($r = 0.062$) and the significance was lowest ($P = 0.350$).

3.2. Thermal benchmarks

3.2.1. Neutral temperature and neutral temperature range

According to ASHRAE Standard ISO-7730 [25], neutral temperature is defined as the temperature at which the human body perceives neither cold nor heat. In this study, the demographic data of the survey respondents, including gender, age, height, weight, clothing insulation, and metabolic rate, along with meteorological parameters such as temperature, humidity, wind speed, and solar radiation, were input into the RayMan model to calculate the Universal Thermal Climate Index (UTCI). From the thermal sensation vote (TSV) data, we calculated the weighted average TSV (MTSV) within each UTCI range (Fig 6(a)). According to ISO 7730, the UTCI value corresponding to TSV = 0 is defined

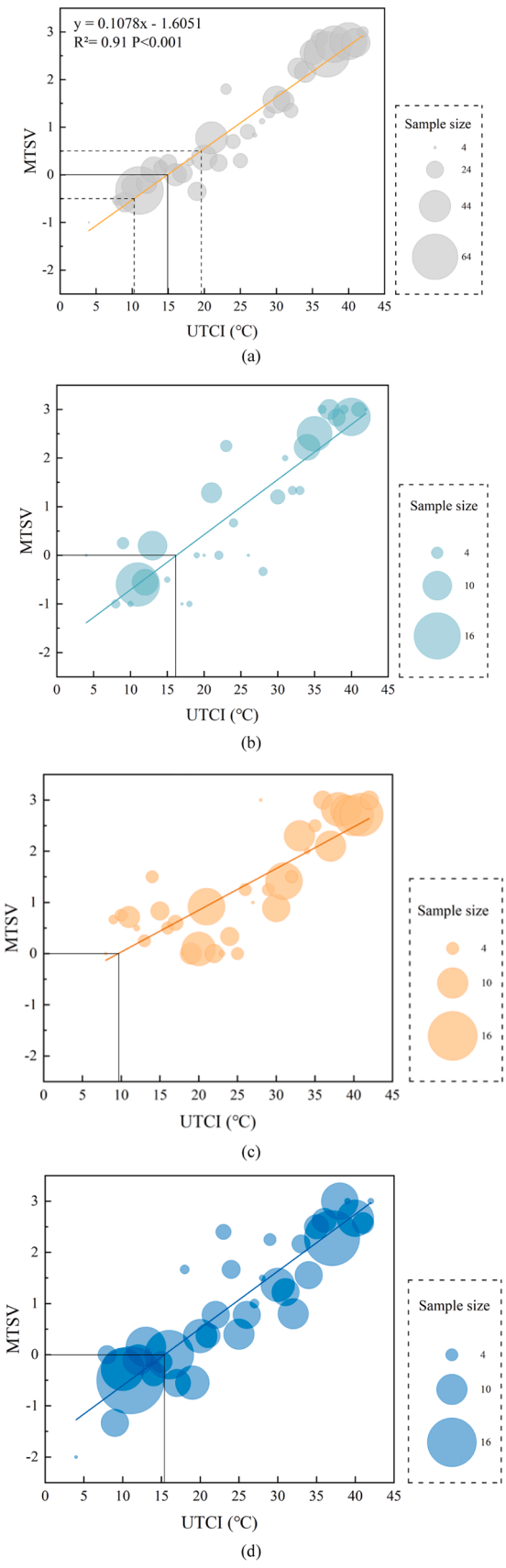


Fig. 6. Correlation between UTCI and MTSV for sound types: (a) Total, (b) Neutral sound, (c) Negative sound, (d) Positive sound.

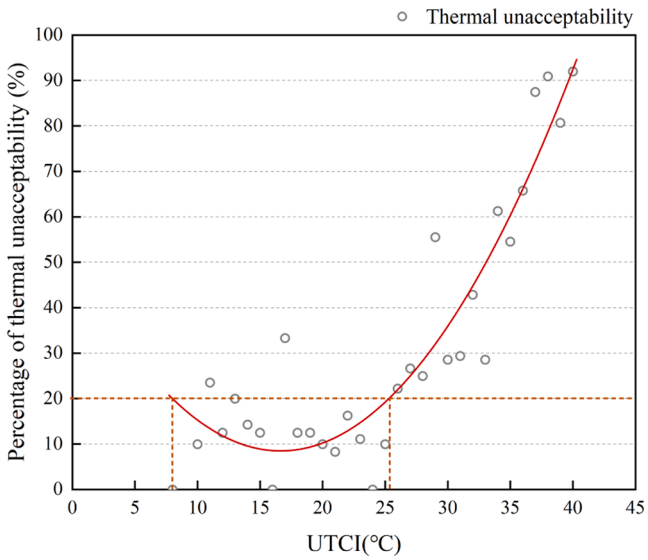


Fig. 7. Relationship between thermal unacceptable rate and UTCI.

as the neutral temperature. The data indicates that the neutral temperature during the survey period was 14.9 °C.

Existing research suggests that the interaction between thermal and acoustic perceptions significantly affects human thermal sensation. Different types of sound environments can influence people's evaluations of their thermal perception to varying degrees. To investigate the impact of different sound types on the thermal comfort of people in urban squares, this study analyzed the neutral temperature range under different sound categories. As shown in Fig 6, under the same UTCI, the average thermal sensation vote in natural sound environments, such as bird songs and water flow, was similar to that in normal environments. The neutral UTCI (NUTCI) under these conditions was 15.9 °C. In contrast, for sound environments dominated by traffic and construction noise, the neutral temperature was 9.4 °C. In neutral sound environments, such as conversations and broadcasts, the neutral temperature was 15.3 °C. This indicates that in natural sound environments, people prefer higher environmental temperatures, increasing their tolerance to thermal conditions. Conversely, in traffic and construction noise environments, people tend to prefer lower temperatures, decreasing their thermal tolerance. In neutral sound environments, respondents' tolerance to temperature was slightly higher than the average neutral temperature.

The temperature range in which MTSV falls between −0.5 and 0.5 is defined as the Neutral Universal Thermal Climate Index Range (NUTCIR) [26]. As shown in Fig 6(a), when TSV is between −0.5 and 0.5, the corresponding NUTCI range is 10.8–19.0 °C. Compared to a study conducted in residential areas of Xi'an (12.4–26.9 °C) [27], the NUTCI range in this study is narrower and lower. This may be due to the study's focus on tourists who have lived in the area for less than six months and are often in a higher state of activity. The annual NUTCIR in Xi'an (12.4–26.9 °C) [28] is relatively wider, which is primarily due to differences in the survey area and respondents. The Xi'an survey was conducted in residential areas where residents are more accustomed to local climatic variations compared to tourists. When respondents mainly heard traffic and construction noise, the neutral UTCI range significantly increased.

3.2.2. *Thermal acceptable range*

ASHRAE Standard 55 defines the Thermal Acceptable Range (TAR) as the temperature range acceptable to at least 80 % (under normal conditions) or 90 % (under stringent conditions) of respondents (ASHRAE 2017) [29]. Conversely, this means that only 20 % (under normal conditions) or 10 % (under stringent conditions) of respondents find the

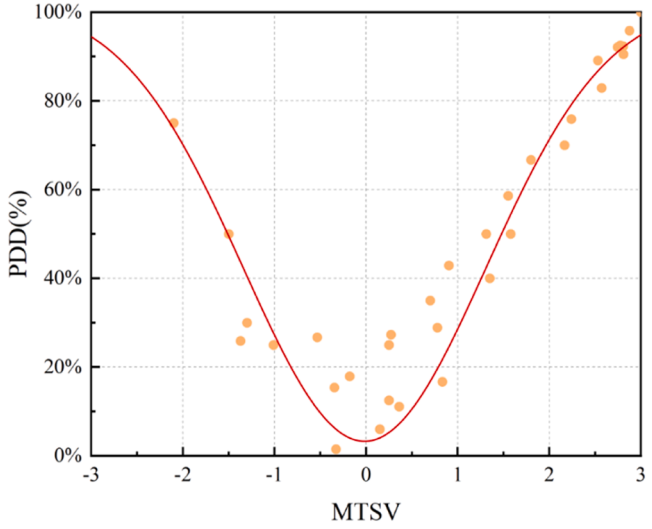


Fig. 8. Correlation between MTSV and PPD.

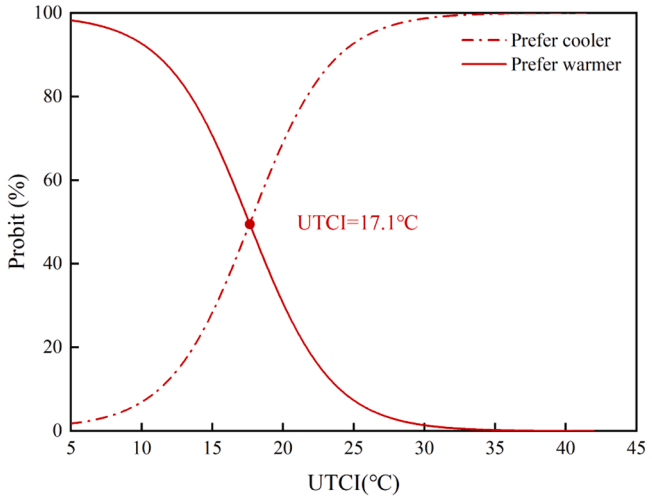


Fig. 9. Preferred UTCI (total).

current thermal environment unsatisfactory. This study adopts the 80 % acceptability criterion to calculate the thermal unacceptable rate within each 1 °C UTCI interval, which is then fitted to a quadratic polynomial (Fig 7). The results indicate that the thermal acceptable temperature range for the surveyed area is 7.8 °C to 25.7 °C, which is closely aligned with the neutral temperature. This suggests that respondents exhibited a high tolerance for uncomfortable temperatures during the survey period.

3.2.3. Percentage of thermal dissatisfaction

ISO 7730 recommends determining the percentage of thermal dissatisfaction (hot/cold) through the Predicted Percentage of Dissatisfied (PPD). In this study, PPD was calculated using TSV data. When TSV ranges from −1 (slightly cool) to +1 (slightly warm), people feel thermally comfortable. However, when TSV ranges from −2 (cool) to −3 (cold) or from +2 (warm) to +3 (hot), people feel thermally uncomfortable [30]. This study calculated the percentage of thermal dissatisfaction for each 1 °C UTCI interval, referred to as the Percentage of Thermal Dissatisfaction, and fitted it with the corresponding MTSV to develop a PPD model for the Xi'an area (Fig 8).

3.2.4. Thermal preference

Preferred temperature is defined as the temperature that people

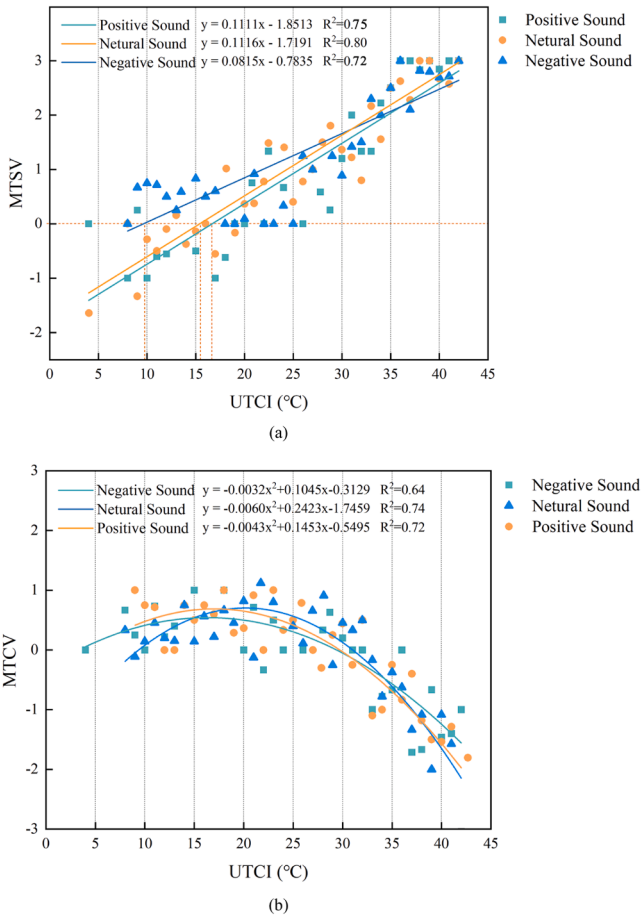


Fig. 10. Correlation between UTCI and MTSV for different sound types(a), Correlation between UTCI and MTCV for different sound types(b).

desire [31], representing their ideal thermal state. The survey recorded the thermal preference votes from respondents at different measurement points. For each 1 °C UTCI interval, we aggregated the votes for "prefer cooler" and "prefer warmer" conditions. The percentages of these votes were calculated and analyzed using probit analysis in SPSS. The intersection of the "prefer cooler" and "prefer warmer" curves represents the preferred UTCI (Fig 9). The analysis shows that the preferred temperature in Xi'an is 17.1 °C, which is 3.6 °C higher than the neutral UTCI (NUTCI). This indicates that respondents in Xi'an tended to prefer higher temperatures during the survey period.

3.3. The impact of thermal and acoustic interactions on human comfort

3.3.1. Classification of thermal and acoustic levels

To analyze the impact of different thermal and acoustic environments on human comfort, this study plotted the data from thermal and acoustic environment votes on SPL-ASV and UTCI-TSV graphs. The thermal and acoustic perceptions were categorized into three levels based on the voting scale. Thermal sensation votes were divided into: cooler (TSV = −3, −2), neutral (TSV = −1, 0, 1), and warmer (TSV = 2, 3). Acoustic sensation votes were categorized into: noisy (ASV = −3, −2), neutral (ASV = −1, 0, 1), and quiet (ASV = 2, 3). Similarly, thermal and acoustic comfort levels were classified into uncomfortable (ACV, TCV = −3, −2), comfortable (ACV, TCV = −1, 0, 1), and very comfortable (ACV, TCV = 2, 3), allowing us to analyze the variation patterns of the thermal and acoustic environments under the influence of a single factor. When studying the impact of different noise types on acoustic perception, noise types are qualitative concepts that lack ordinality. Therefore, sounds were categorized based on noise types into

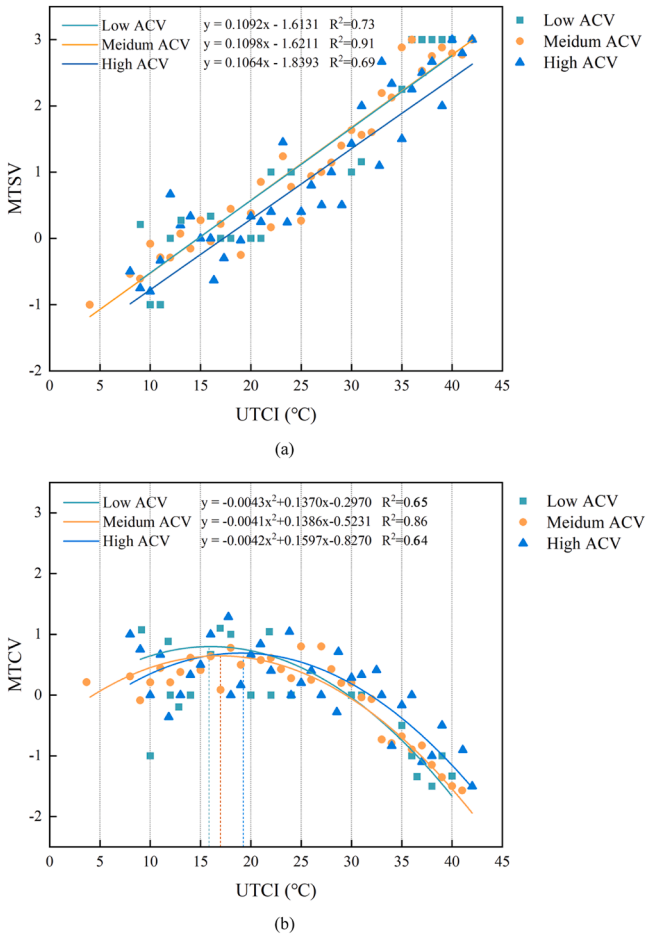


Fig. 11. (a) Correlation between UTCI and MTSV for different ACV, (b) Correlation between UTCI and MTCV for different ACV.

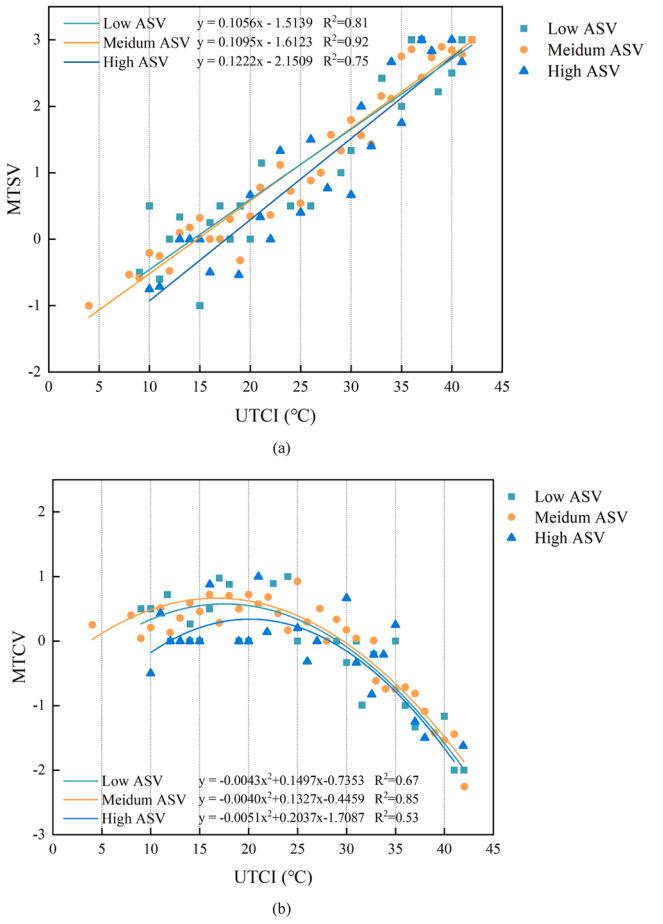


Fig. 12. (a) Correlation between UTCI and MTSV for different ASV, (b) Correlation between UTCI and MTCV for different ASV.

positive sound sources, neutral sound sources, and negative sound sources. Through comparative analysis, this study explored the influence of sound types on thermal and acoustic comfort.

3.3.2. *Impact of acoustic environment on thermal perception*

The mean thermal sensation votes (TSV) corresponding to each 1 °C UTCI interval were plotted in the same coordinate system and fitted with a quadratic polynomial to derive the variation pattern of TSV with UTCI (Fig 10a). The graph illustrates that irrespective of the type of sound, TSV gradually increases with rising Universal Thermal Climate Index (UTCI). In negative acoustic environments, respondents exhibit a lower slope in thermal sensation voting ($k = 0.082$), whereas under neutral ($k = 0.111$) and positive sound conditions ($k = 0.111$), the slopes are closer. At higher temperatures, the type of sound at measurement points has no significant impact on thermal sensation vote. However, at lower temperatures (UTCI < 30 °C), respondents exposed to negative acoustic environments significantly increase their voting values for current thermal conditions. When respondents perceive the environment as neither cold nor warm (TSV=0), those affected by traffic and noise tend to prefer lower temperatures (9.6 °C), whereas positive sound environments lead people to prefer warmer environments (17.2 °C). This suggests that poorer acoustic conditions can induce irritation and psychological discomfort, prompting respondents to desire lower environmental temperatures.

Fig 10(b) illustrates the variation of thermal comfort votes (TCV) with UTCI. During the survey period, respondents generally felt warmer, resulting in a temperature distribution mainly concentrated in the 20–40 °C range. The results indicate that within the comfortable temperature range, positive sound environments expand the UTCI range where people feel comfortable. In uncomfortable thermal environments, respondents exposed to positive sound sources have the highest TCV values.

3.3.3. *Impact of acoustic comfort on thermal perception*

Fig 11(a) demonstrates that regardless of changes in acoustic comfort, the slope of human thermal sensation remains relatively stable ($0.106 < k < 0.110$) with temperature variation. When acoustic comfort is higher (ACV=2,3), respondents generally rate their thermal sensation lower. Thus, within the measurement points, higher acoustic comfort levels lead to a slight decrease in thermal sensation. Across different conditions, the slope of thermal sensation curves remains consistent, indicating that acoustic comfort does not alter the sensitivity of individuals to thermal environments.

The variation of human thermal comfort with temperature under three acoustic comfort levels is depicted in Fig 11(b). In the curve of thermal comfort versus temperature, the X-axis value corresponding to the apex of the curve is the most comfortable temperature. From the figure, the lowest X-axis vertex value is found in the low acoustic environment condition (15.9 °C), followed by the second highest X-axis value in the meidum acoustic environment (16.9 °C), and the highest X-axis value in the high acoustic environment (19.1 °C). This indicates that respondents tended to prefer lower temperatures in low acoustic environments. Although this section was conducted with respondents in the cold region of Xi'an, the study covers a wide range of temperatures, so the effect of acoustic comfort on changes in thermal perception is still relevant in multiple regions. The optimization of the built environment as well as the improvement of the thermal environment plays an

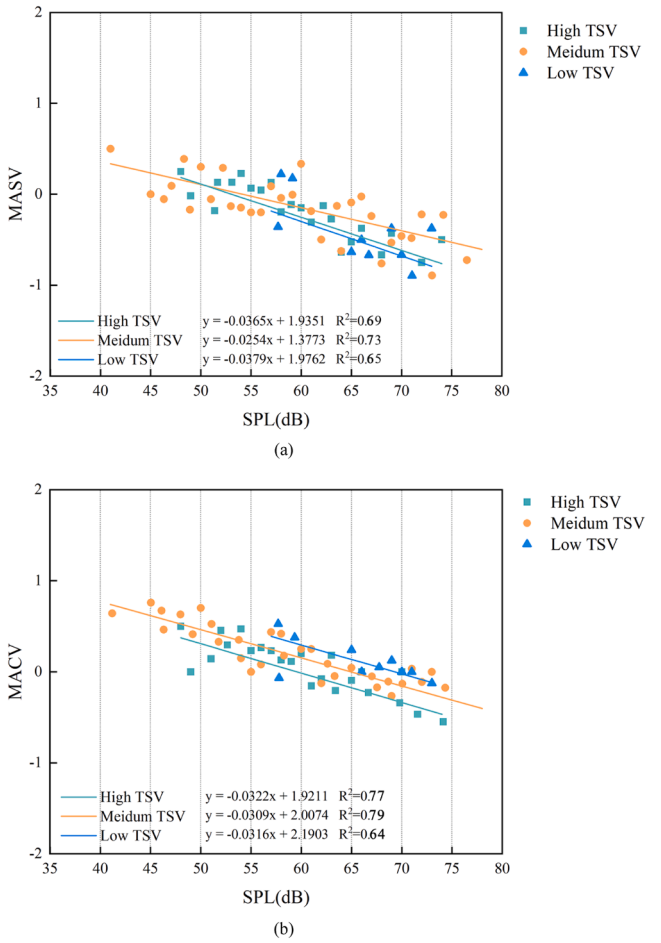


Fig. 13. (a) Correlation between SPL and MASV for different TSV, (b) Correlation between SPL and MACV for different TSV.

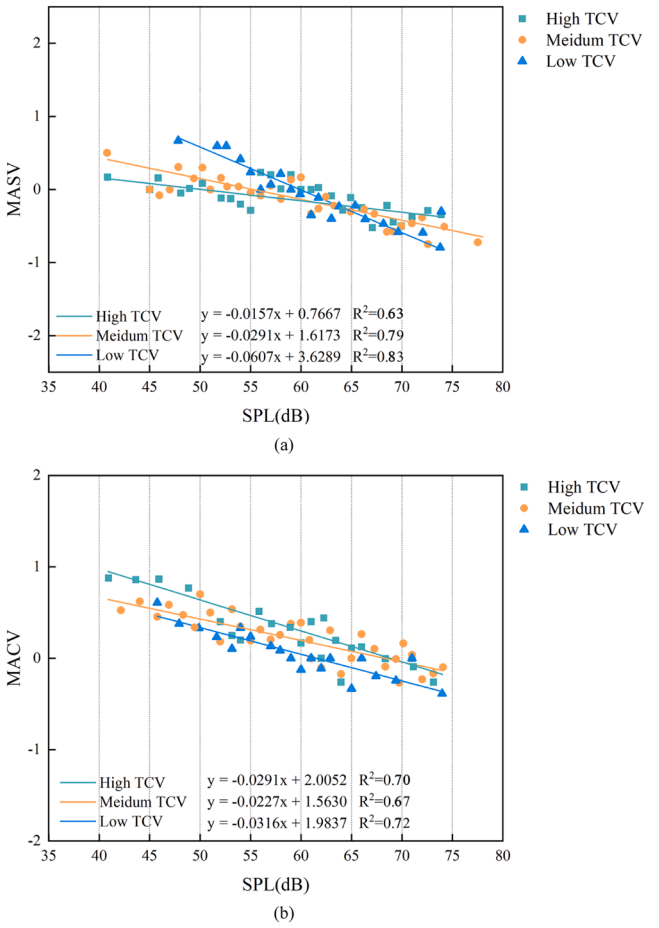


Fig. 14. (a) Correlation between SPL and MASV for different TCV, (b) Correlation between SPL and MACV for different TCV.

important role from the point of view of energy management and public health and safety, and Koley et al. [32] have shown that a good indoor and outdoor thermal environment is essential for predicting and controlling the spread of diseases.

3.3.4. Impact of sound pressure levels on thermal perception

Fig 12(a) illustrates that in quiet environments, the slope of thermal sensation with UTCI variation (k = 0.122) is higher compared to moderate (k = 0.110) and noisy environments (k = 0.106). As the sound sensation votes increase, these differences gradually diminish. The results indicate that in quieter acoustic environments (ASV=2,3), individuals are more sensitive to changes in thermal environments. In cooler conditions, a quieter acoustic environment (ASV=2,3) tends to lower thermal sensation. This may be attributed to sound acting as energy in open spaces, consistent with existing findings in acoustic environment research [33].

At higher environmental temperatures, the thermal comfort index for individuals shows no significant variation across the three acoustic conditions (Fig 12(b)). Respondents in quiet environments (ASV=2,3) perceive the highest comfort temperature, followed by noisy environments (ASV=−2,−3), with moderate acoustic environments (ASV=−1,0,1) perceived as least comfortable. This aligns with previous research conclusions that both noisier and more negative noise sources increase respondent irritation, reducing their perceived comfort temperature. The moderate acoustic environment showed the largest TCV at the most comfortable thermal temperature (0.64), followed by the noisy environment (TCV of 0.49), and the smallest TCV extremes in the quiet environment (0.37). This suggests that the discomfort of the acoustic environment diverted the respondents' attention from thermal discomfort to some extent, which is consistent with the findings of previous studies.

Table 5
NUTCI and NUTCIR among OTC studies.

Author	Climate zone	City, country	NUTCI (°C)	NUTCIR (°C)
This study	Cwa	Xi'an, China	14.9	10.8–19.0
Huang et al. [35]	Cwa	Xi'an, China	13.9	6.4–21.5
Hadianpour et al. [36]	BWk	Tehran, Iran	14.3	14.2–20.1
Shao et al. [37]	Dwa	Harbin, China	6.7	0.5–14.0
Feng et al. [38]	Cfa	Guangzhou, China	23.7	21.3–26.1
Guo et al. [39]	Dwa	Dalian, China	18.8	14.7–23.0

3.3.5. Influence of thermal environment on sound perception

Regardless of changes in the thermal environment, sound pressure level was the most important factor influencing respondents' sound perception vote. Meanwhile, AL et al. chose a university campus to evaluate the combined effect of thermal and acoustic environments on human comfort and concluded that the interaction of TCV, ASV or ACV is not significant. [34] This is consistent with the findings in this section.

From Fig 13(a), it is observed that when the current thermal environment is cold (TSV = −3,−2) or warm (TSV = 2,3), the slope of sound sensation with changes in sound pressure level (k = 0.038, 0.037) is higher compared to moderate thermal sensation (k = 0.025). Under cold

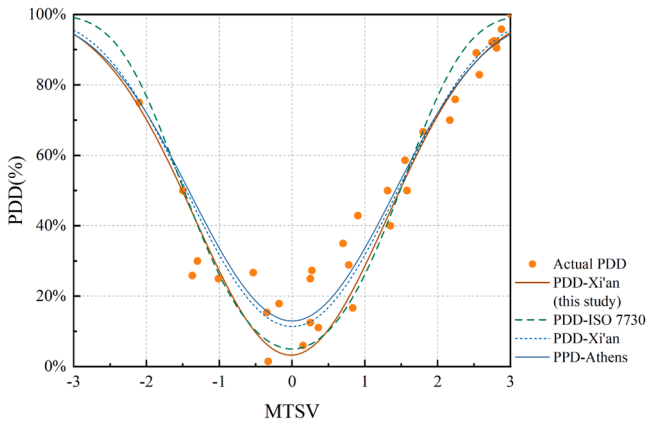


Fig. 15. Analyses of PPD.

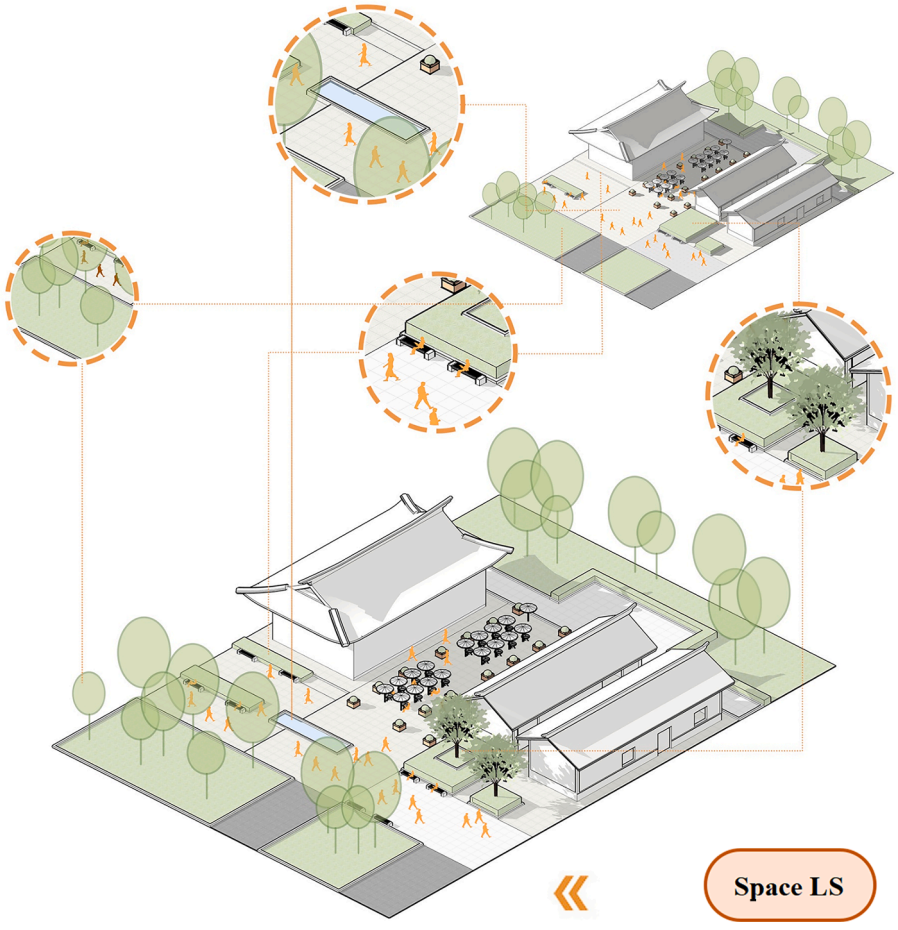
conditions, respondents' perception of sound sensation is slightly lower than in warm environments. This indicates that at comfortable environmental temperatures, sensitivity to changes in sound pressure level decreases, suggesting that appropriate air temperature can increase tolerance to higher sound pressure levels. In all three TSV conditions, ASV decreased with increasing sound pressure level.

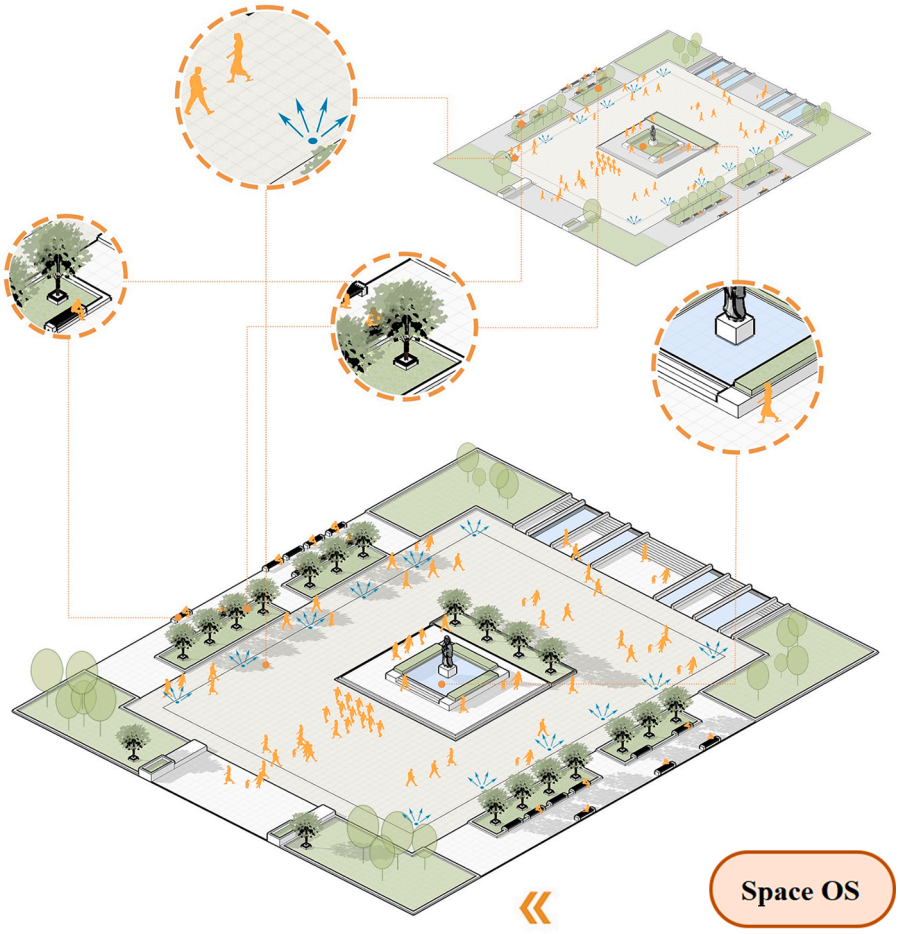
Regardless of variations in respondents' thermal environment conditions, the slope of sound comfort influenced by noise sound pressure level is similar (k = 0.032, 0.031, 0.032) (Fig 13(b)). When the thermal environment is cold (TSV=−2,−3), overall sound comfort is highest. Moderate environments follow, with the lowest sound comfort in warm environments. This suggests that as respondents' thermal sensation increases, overall sound comfort decreases. Specifically, within a moderate range of sound comfort, a cold thermal environment increases tolerance to higher sound pressure levels.

3.3.6. *Influence of thermal comfort on sound perception*

In terms of sound sensation, respondents' perception varies significantly with different levels of thermal comfort as depicted in Fig 14(a). The distribution of points in this chart is more scattered due to the lower impact of the thermal environment on sound perception and the fact that the current sound pressure level corresponds to a number of different sound types. When respondents are in a comfortable state (TCV=2,3), the slope of the fitted curve with SPL is lowest (k = 0.016), while it is highest when thermal conditions are uncomfortable (TCV=−3,−2), reaching 0.061. At an ambient sound pressure level of 60~65 dB, the effect of thermal comfort on acoustic perception is relatively low. This is mainly due to the fact that the acoustic sensation remains stable in this sound pressure level range, regardless of thermal comfort changes, the ASV values are centered between −0.33 and 0.24. These results indicate that a comfortable thermal environment effectively reduces respondents' sensitivity to sound levels, enhancing their tolerance to sound pressure levels. Therefore, in outdoor environments with high foot traffic and frequent high sound pressure levels, creating a reasonable thermal environment can promote perceived comfort in sound environments.

When thermal comfort voting is moderate, the slope of respondents' sound comfort change (k = 0.023) is lower than in comfortable (k = 0.029) and uncomfortable (k = 0.032) thermal environments (Fig 14 (b)). Overall, at the same sound pressure level, the acoustic comfort of the respondents increased with increasing levels of thermal comfort. A





comfortable thermal environment contributes to acoustic comfort.

4. Discussion

4.1. Outdoor thermal indices

4.1.1. Neutral temperature

This study is conducted in Xi'an, a cold region. To compare thermal comfort variations under different conditions and with overall population data, several studies from different regions were used to extract neutral temperatures. Xi'an's neutral UTCI is similar to year-round surveys conducted in the same region [35] (Table 5), being 0.7 °C higher than Tehran [36]. This similarity indicates similar thermal perceptions among residents in these areas. Xi'an's neutral UTCI is lower than Athens [34] by 5.4 °C and Guangzhou [38] by 8.8 °C, but higher than Harbin [37] by 8.2 °C. These differences in NUCTI are due to climatic variations: Athens has a warmer climate, Guangzhou is in a subtropical region, and Harbin experiences severe cold.

4.1.2. Range of neutral temperature

The NUTCIR observed in this study is narrower compared to year-round NUTCIR in the same surveyed area, wider than Tehran and Guangzhou, and closer to Dalian. This variation may stem from the study's focus on children in Xi'an [35], who exhibit stronger adaptive capabilities to thermal changes. In contrast, Tehran [36] experiences more stable climatic conditions. Residents in Xi'an are thus more adaptable to the diverse seasonal changes in thermal environments. The lower limit of NUTCIR in this study is higher than Harbin [37] by 10.3 °C, while the upper limit is lower than Athens [34] by 5.2 °C and Dalian [39] by 4.0 °C. This indicates varying levels of adaptation to thermal environments among residents living in different climatic zones.

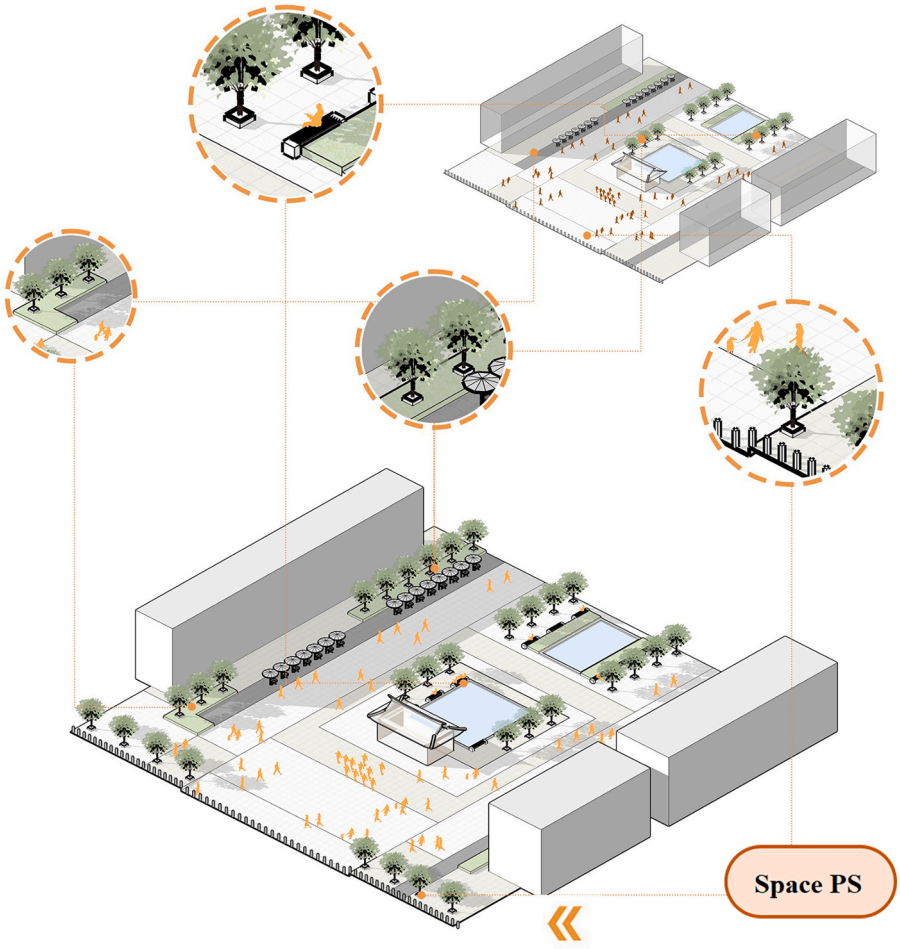
4.1.3. Percentage of thermal discomfort

When MTSV = 0, the PPD in Xi'an's campus is lower compared to other surveyed areas within the city (Fig 15). This suggests that during the study period, respondents in this area had a higher tolerance for their environment. Within discomfort zones (MTSV = +2, +3) or (MTSV = −3, −2), Xi'an's PPD is lower than ISO 7730 standards, indicating that the population in Xi'an is more capable of adapting to uncomfortable temperatures compared to the standard.

4.2. Strategies for enhancing thermal-acoustic comfort in urban open spaces

4.2.1. Physical environment status of open squares

This study highlights the significant influence of physical factors on the thermal-acoustic environment of open square spaces. Targeted landscaping can effectively improve outdoor micro-climates. In the LS site, globe radiation is significantly lower compared to other spaces. This area exhibits higher thermal comfort in summer but lower comfort in winter due to lower temperatures. The OS site, being open and unobstructed, experiences higher solar radiation in summer and reduced thermal sensation in winter due to high wind speeds. Moreover, this space has the lowest relative humidity, exacerbating thermal discomfort. The PS measurement point features abundant water features that increase humidity, alleviating thermal discomfort, while large trees on the west side provide ample shaded areas, reducing solar radiation in summer. The SS site, enclosed by high walls on both sides, experiences



the lowest average wind speed and overall higher comfort.

In terms of acoustic environment, the LS site, located at the end of a road, has the lowest average sound pressure level (SPL Mean = 58 dB), followed by the SS site (SPL Mean = 59.29 dB). The OS site, close to traffic flow and gatherings, exhibits the highest average sound pressure level (SPL Mean = 63.26 dB). Similarly, the PS site near road traffic also experiences an average noise level above 60 dB (SPL Mean = 60.02 dB). Existing research indicates that sound pressure levels exceeding 60 dB can induce discomfort in perception, increasing psychological agitation. Therefore, measures should be implemented at OS and PS sites to reduce sound pressure levels and improve acoustic conditions.

4.2.2. Design strategies for square environment optimization

Existing studies have found that when people enter a foggy environment, water droplets cause convection and evaporative heat loss on the skin surface, reduce skin temperature (ST) and relieve thermal discomfort. [40] On this basis, Su et al. [41] analyzed the effects of the setting height of the spray device and the amount of spray on the human body in different types of environments. Therefore, for space OS and space SS, a spray device is added in summer to increase humidity, alleviate the thermal discomfort caused by solar radiation, reduce the temperature of the human surface through water vapor evaporation, and increase the overall comfort of the environment.

There are many kinds of materials for landscape sketches, and too high temperature may cause thermal discomfort or even scalding, which will affect people's safety. Huang et al. [35] studied the influence of outdoor landscape and seat surface material on the thermal perception of the human body, and concluded that the temperature of some landscape sketches is higher when exposed to direct sunlight, which seriously affects the thermal safety of children.

To alleviate the discomfort of the heat, people who chose to rest in the shade voted the most. For the OS, SS and PS spaces that are exposed to direct sunlight in summer, it is expected to set up sunshade facilities in combination with landscape sketches, pay attention to public participation and interaction, and expect to achieve artistic aesthetics and practical application value.

Optimizing the environmental conditions of measurement point spaces should be based on empirical data, using layout and landscape composition to strategically improve thermal-acoustic environments and enhance overall comfort in outdoor activity spaces.

Problems

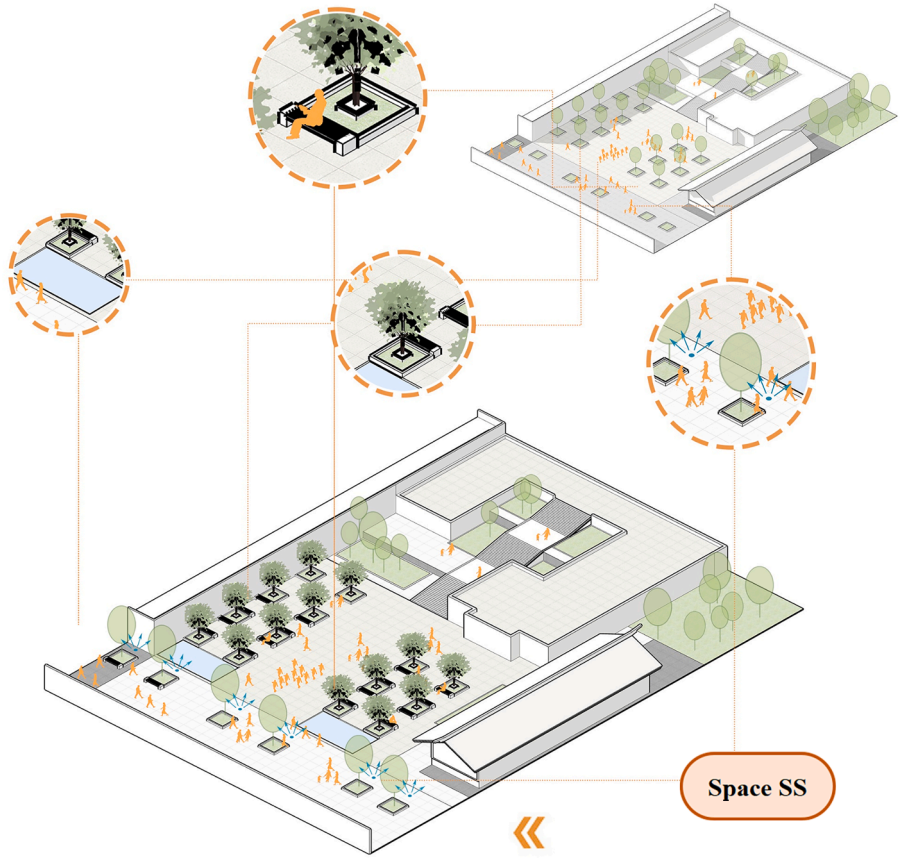
1. Inadequate shading facilities.
2. High solar radiation in summer.
3. Low site humidity and lack of vegetation.

Strategies

1. Introducing mobile shading devices.
2. Planting tall deciduous trees on the west side of the site.
3. Installing landscaped pools and low shrubs.

Problems

1. Inadequate tree cover and vegetation, leading to high solar radiation.



2. Low site humidity, resulting in poor heat dissipation during summer.
3. Limited number of resting spaces.

Strategies

1. Planting vegetation on both sides of the square to provide shade.
2. Installing misting devices to expand the distribution of water features.
3. Adding sunshades, pavilions, or benches for resting facilities.

Problems

1. Inadequate resting spaces on both sides of the square.
2. Poor acoustic environment near the roads.
3. Single-layered paving materials prone to heat accumulation in summer.

Strategies

1. Installing sunshades, pavilions, or benches for resting facilities.
2. Designing sound isolation barriers or green plant belts.
3. Installing misting devices to enhance rainwater penetration.

Problems

1. High wind speeds and cold winters.
2. Few trees leading to low humidity and insufficient summer shade.
3. High pedestrian traffic with single-layered paving materials in walkways, resulting in heat accumulation in summer.

Strategies

1. Planting tall deciduous trees to reduce wind speed.
2. Installing ponds within the site to provide summer shade.
3. Installing misting devices along the roadside to lower the perceived temperature for pedestrians.

4.3. Limitations and future work

Our study has several limitations. 1) the acoustic environment parameters at measurement points may exhibit significant fluctuations, which are less stable compared to meteorological parameters. For future study we should continue to monitor the acoustic environment to improve the accuracy of the research. 2) this study only considered respondents' thermal and acoustic perceptions; future research should delve deeper into other factors such as light.

5. Conclusion

In an open square space in Xi'an, four typical spaces were selected for systematic study and comparison of thermal and acoustic physical environments and environmental landscape factors influencing people's thermal perception evaluations outdoors, using methods such as meteorological measurements and questionnaires. A series of strategies for multidimensional optimization of the square were proposed. The conclusions are as follows:

1. The UTCI of Xi'an urban squares is 14.9 °C, with a neutral UTCI range of 10.8~19.0 °C, preferred temperature of 17.1 °C, and thermal comfort range of 7.8 °C-25.7 °C.
2. When thermal sensation votes are neutral (TSV=−1,0,1), respondents influenced by negative sound tend to prefer lower

temperatures, while under positive sound types, people perceive lower thermal sensations. Negative sound environments increase discomfort among respondents at lower temperatures (UTCI < 15 °C).

3. When the acoustic environment is quieter (ASV=2,3), people are more sensitive to changes in thermal environment perception. Appropriate environmental sound pressure levels (ASV=−3,−2,−1,0,1) can mitigate thermal discomfort under cooler conditions (UTCI < 25 °C) to a certain extent. As the thermal environment increases, people's acoustic comfort decreases.
4. When respondents are in a comfortable state (TCV=2,3), the average slope of acoustic sensation is minimized, whereas in uncomfortable thermal conditions (TCV=−3,−2), the slope is maximized, indicating that a comfortable thermal environment can effectively reduce respondents' sensitivity to environmental sound pressure levels.
5. Based on meteorological parameter characteristics and respondent voting for different square spaces, optimized designs were provided. To enhance thermal comfort in the four typical spaces, methods such as installing cooling mist devices and optimizing tree species along walkways can be employed to improve microclimate comfort.

Declaration

There are no conflicts of interest to declare.

CRediT authorship contribution statement

Meng Zhen: Conceptualization, Methodology, Writing – review & editing, Supervision. **Xinyi Yuan:** Writing – original draft, Data curation, Visualization. **Liqiang Gao:** Project administration, Resources. **Weijie Xia:** Writing – review & editing, Formal analysis, Software, Visualization. **Kai Nan:** Investigation, Project administration, Resources, Validation. **Ming Zhang:** Investigation, Project administration, Resources, Validation.

Declaration of competing interest

None.

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Supplementary materials

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